# NHS Outpatient Attendance and DNA Performance Analysis

## 1 PROJECT OVERVIEW

This project analysed NHS England outpatient attendance data for the 2025/26 reporting period using Microsoft Excel, SQLite and Power BI.

The aim was to calculate did-not-attend (DNA) rates for healthcare providers, investigate differences in appointment attendance and present the results through an interactive dashboard.

The project followed a complete analytical workflow, from exploring and cleaning the original dataset to performing SQL calculations, validating results and developing visualisations.

## 2 DATASET

The analysis used the NHS England Hospital Outpatient Activity 2025/26 dataset.

The original dataset contained:

- 55,845 records
- 1,819 distinct provider codes
- 21 attendance-related measures
- 7 columns

The seven columns were:

| Column | Description |
|---|---|
| REPORTING_PERIOD | Reporting period |
| GEOGRAPHY_LEVEL | Reporting level |
| ORGANISATION_CODE | Provider identifier |
| ORGANISATION_DESCRIPTION | Provider name |
| MEASURE_TYPE | Category of measure |
| MEASURE | Specific attendance measure |
| MEASURE_VALUE | Reported activity count |

The dataset used a long format, meaning each provider's attendance measures appeared in separate rows.

The analysis focused on provider-level records (03.Provider) within the Attendance Type category.

Four measures were selected:

1. 01.Attended first appointment
2. 03.Attended subsequent appointment
3. 06.Did not attend first appointment
4. 08.Did not attend subsequent appointment

These measures formed the basis of the selected DNA rate.

# 3 DATA EXPLORATION AND CLEANING IN EXCEL

Microsoft Excel was used for initial data exploration, cleaning, calculations and quality checks.

The source data was organised into an Excel table named tblOutpatients. Filters, formulas and pivot tables helped examine provider records, measure availability and attendance activity.

The workbook contained seven worksheets

| Worksheet | Purpose |
|---|---|
| Working_Data | Source records and working calculations |
| Pivot | Pivot-table analysis |
| KPI_Summary | Summary performance indicators |
| Provider_Comparison | Provider-level calculations |
| Top_10_DNA | Highest DNA rates |
| Data_Quality | Data-quality checks |
| Measure_Audit | Attendance measure assessment |

## 3.1 DATA QUALITY RULES

The source contained suppressed values represented by *. These values were not treated as zero because their actual counts were unavailable.

Providers needed one unique, valid numeric record for each of the four selected measures.

A minimum threshold of 1,000 selected appointments was also applied to reduce the influence of very small appointment volumes on provider comparisons.

The threshold was an analytical decision, not an official NHS requirement.

After applying the inclusion criteria, 469 providers qualified for the final analysis.

# 4 CALCULATING DNA RATES

Four main metrics were calculated for each qualifying provider.

**Attended appointments**

Attended first appointments + Attended subsequent appointments

**DNA appointments**

DNA first appointments + DNA subsequent appointments

**Selected appointment volume**

Attended appointments + DNA appointments

**Selected DNA rate**

(DNA appointments ÷ Selected appointment volume) × 100

For example, Higher Blackley Medical Centre recorded:

| **Measure** | **Count** |
|---|---|
| Attended first appointments | 925 |
| Attended subsequent appointments | 390 |
| DNA first appointments | 205 |
| DNA subsequent appointments | 75 |
| **Total attended** | **1,315** |
| **Total DNA** | **280** |
| **Selected appointment volume** | **1,595** |

Its selected DNA rate was:

**280 ÷ 1,595 × 100 = 17.55%**

This calculation was applied consistently across the qualifying providers.

# 5 SQL Analysis

The imported table was named outpatient_activity.

SQL was used to reproduce the provider-level calculations more systematically.

The analysis involved:

- WHERE to filter provider-level records and the four selected measures.
- CASE to separate attended appointments from DNAs.
- SUM to calculate appointment totals.
- COUNT and measure checks to assess completeness.
- GROUP BY to aggregate records by provider.
- Numeric conversion and filtering to exclude invalid values.
- Inclusion rules to retain providers with at least 1,000 selected appointments.

A SQL view named vw_provider_dna_analysis was created.

The view produced six columns:

| Column | Description |
|---|---|
| provider_code | Provider identifier |
| provider_name | Provider name |
| attended | Total selected attended appointments |
| dna | Total selected DNAs |
| appointment_volume | Total selected appointments |
| selected_dna_rate_pct | Selected DNA rate (%) |

The view returned **469 qualifying providers**.

The results were exported to nhs_provider_dna_analysis.csv for Power BI.

The SQL code was also saved separately as provider_dna_analysis.sql.

# 6 DATA VALIDATION

Validation was performed to check consistency between Excel and SQLite.

- The main checks confirmed:
- Both approaches identified 469 qualifying providers.
- The Top 10 provider rankings matched.
- The Top 10 selected DNA percentages matched.
- Higher Blackley Medical Centre had the highest selected DNA rate at 17.55%.

The SQL results were also checked against the Power BI dashboard.

One limitation remained in the Excel data-quality audit: an unresolved classification discrepancy affecting 144 excluded providers.

This discrepancy concerned the classification of excluded records rather than the independently validated qualifying-provider count. A complete reconciliation of all excluded-provider categories was not finished.

# 7 POWER BI DASHBOARD DEVELOPMENT

The SQL output was imported into Power BI as the Provider_DNA_Analysis table.

Three DAX measures were created to calculate the main performance indicators:

- **Qualifying Providers:** Distinct count of provider codes.
- **Selected Appointments:** Sum of selected appointment volumes.
- **Weighted DNA Rate:** Total DNA appointments divided by total selected appointments.

The weighted calculation was important because providers had different appointment volumes. Taking the simple average of provider DNA percentages would give smaller and larger providers equal influence.

The weighted rate instead reflects the proportion of selected appointments missed across the qualifying providers.

### 7.1 Dashboard Visualisations

The final dashboard included:

- **KPI cards:** Displayed the 469 qualifying providers, approximately 99 million selected appointments and a 7.06% weighted DNA rate.
- **Top 10 bar chart:** Ranked the ten qualifying providers with the highest selected DNA rates.
- **Scatter chart:** Compared provider appointment volumes with selected DNA rates to explore variation between providers of different sizes.
- **DNA rate distribution chart:** Grouped providers into percentage bands, including 0–2%, 2–4% and subsequent two-percentage-point ranges. A numeric sorting column ensured the bands appeared in the correct order.
- **Provider comparison table:** Displayed detailed provider-level results.
- **Searchable provider slicer:** Allowed users to select an individual provider and explore its results.
- A methodology footer explained the selected measures, suppression handling and minimum-volume threshold.

## 8 Key Results

| Metric | Result |
|---|---|
| Source records | 55,845 |
| Distinct source provider codes | 1,819 |
| Qualifying providers | 469 |
| Selected appointments | Approximately 99 million |
| Weighted selected DNA rate | 7.06% |
| Highest selected DNA rate | 17.55% |
| Highest-rate provider | Higher Blackley Medical Centre |

### 8.1 Top 10 providers by selected DNA rate

| Rank | Provider | DNA rate |
|---|---|---|
| 1 | Higher Blackley Medical Centre | 17.55% |
| 2 | North Staffordshire Combined Healthcare NHS Trust | 17.25% |
| 3 | Sussex Partnership NHS Foundation Trust | 16.94% |
| 4 | Specsavers Healthcare London Greenford Broadway | 15.61% |
| 5 | The Westbourne Centre | 15.31% |
| 6 | Park Way House | 12.86% |
| 7 | Longsight Medical Practice | 12.75% |
| 8 | Leicestershire Partnership NHS Trust | 12.68% |

| | | |
|---|---|---|
| 9 | Barking Havering and Redbridge University Hospitals NHS Trust | 12.66% |
| 10 | Birkenshaw Health Centre | 12.27% |

# 9 FINDINGS AND INSIGHTS

The analysis identified considerable variation in selected DNA rates across qualifying providers.

The overall weighted rate was 7.06%, while the highest-ranking provider recorded 17.55%.

This difference highlights the importance of examining individual providers rather than relying exclusively on an overall figure.

Appointment volume also matters when interpreting DNA performance. Providers with similar percentages might record substantially different numbers of missed appointments.

The scatter chart enabled comparisons between appointment volume and DNA rate, while the distribution chart provided a broader view of provider variation beyond the Top 10.

The results identify organisations and patterns that could warrant further investigation. They do not establish why appointments were missed or whether differences reflect provider performance, patient characteristics or service arrangements

# 10 LIMITATIONS

The analysis has several important limitations.

**Selected measures:** Only four attendance measures were included. The 7.06% figure is not the official overall NHS outpatient DNA rate.

**Provider exclusions:** The final analysis covered 469 qualifying providers rather than all 1,819 source provider codes.

**Volume threshold:** The 1,000-appointment threshold was chosen for this project. A different threshold could change the results.

**Provider differences:** The organisations provide different services and serve different patient populations. The rankings are descriptive, not risk-adjusted comparisons.

**Single reporting period:** The project examined 2025/26 activity and did not assess changes over time.

**No causal analysis:** The dataset did not provide sufficient information to explain the reasons for missed appointments.

**Data-quality audit:** The Excel audit contained an unresolved classification discrepancy affecting 144 excluded providers, although the main qualifying-provider count and Top 10 results were independently validated.

# 11 PROJECT WORKFLOW SUMMARY

The complete workflow was:

1. **Data acquisition:** Obtained the NHS England outpatient activity dataset.
2. **Excel exploration:** Inspected records, measures and provider activity.
3. **Data cleaning:** Checked suppressed values, missing measures, numeric validity and completeness.
4. **Metric definition:** Selected four attendance measures and established the DNA rate formula.
5. **Provider filtering:** Applied the 1,000-appointment threshold and identified 469 qualifying providers.
6. **SQL analysis:** Aggregated provider-level activity and created a reusable SQL view.
7. **Validation:** Compared the main Excel and SQL outputs.
8. **Data export:** Exported the SQL results to CSV.
9. **Power BI development:** Created DAX measures, charts, tables and interactive filtering.
10. **Documentation:** Recorded the methodology, findings and limitations.

# 12 CONCLUSION

This project transformed 55,845 NHS outpatient activity records into a provider-level analysis and interactive Power BI dashboard.

Using Excel, SQLite and Power BI, the analysis identified 469 qualifying providers and calculated a weighted selected DNA rate of 7.06% across approximately 99 million selected appointments.

The dashboard highlighted variation between providers, with Higher Blackley Medical Centre recording the highest selected DNA rate of 17.55%.

The project demonstrates practical skills in data exploration, cleaning, SQL aggregation, validation, DAX calculations and dashboard development.

Its main outcome is a reproducible analytical workflow and a clear reporting tool for examining selected outpatient attendance activity, with transparent assumptions and documented limitations.